

Lab 9: Fast Fourier Transform (FFT)

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1 Introduction

The objective of this laboratory was the empirical verification of the Fast Fourier Transform (FFT) for signal analysis and noise reduction using the GNU Scientific Library (GSL). [cite_start]The Discrete Fourier Transform (DFT) is an essential mathematical tool for decomposing a time-domain signal into its constituent frequency components[cite: 21]. [cite_start]This report details the complete experimental process, including multi-component signal generation, spectral analysis, and the application of frequency-domain filtering to remediate stochastic noise[cite: 21].

2 Methodology

[cite_start]The computational procedure was structured into six distinct tasks as defined in the laboratory requirements[cite: 17, 21]:

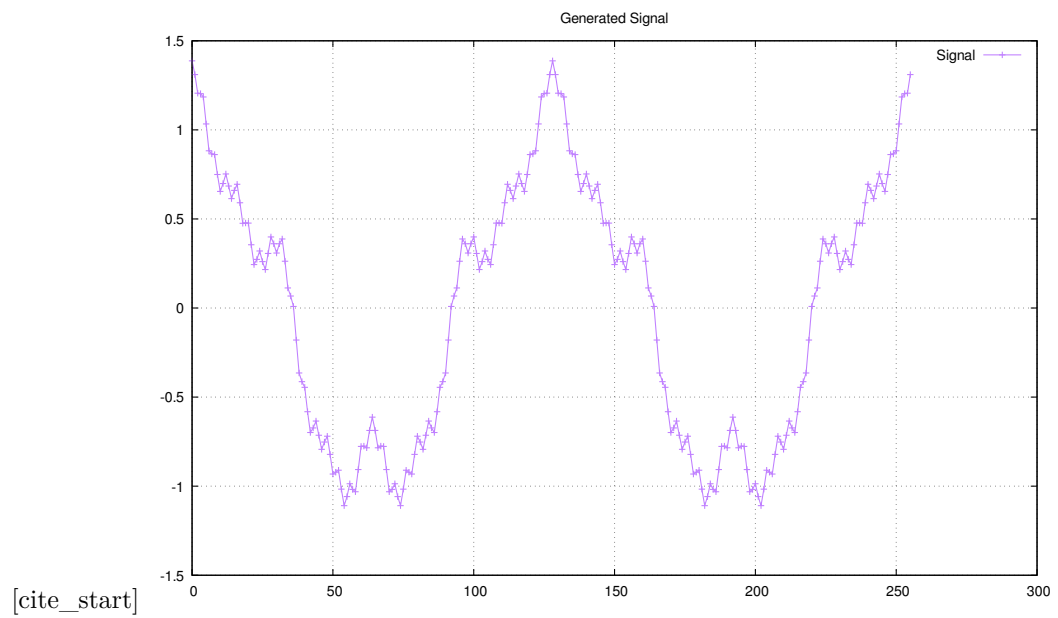
1. [cite_start]**Signal Generation:** Synthesis of a 256-element array representing a sum of four cosine functions with varying periods ($k = 2, 8, 16, 64$) and amplitudes $(1, 1/5, 1/8, 1/16)$ [cite: 17, 21].
2. [cite_start]**Spectral Analysis:** Extraction of the frequency spectrum using the `gsl_fft_real_radix2_transform` routine[cite: 17, 18, 21].
3. [cite_start]**Noisy Signal Simulation:** Generation of a cosine wave corrupted by random white noise using a standard pseudo-random number generator[cite: 17, 21].
4. [cite_start]**Noise Spectrum Analysis:** FFT application to visualize the wide-band frequency distribution of the interference[cite: 17, 21].
5. [cite_start]**Frequency Filtering:** Implementation of a magnitude-based threshold filter to zero out spectral components with an absolute value below 50[cite: 17, 21].
6. [cite_start]**Inverse Transform:** Signal reconstruction using the `gsl_fft_halfcomplex_radix2_inverse` function to evaluate filtering efficiency[cite: 17, 18, 21].

3 Results

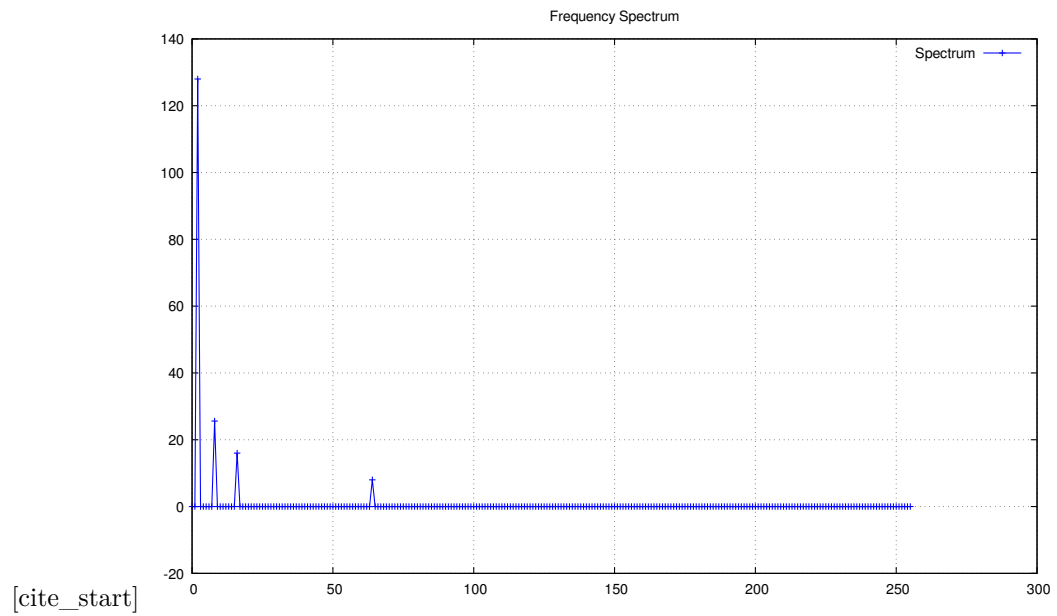
3.1 Signal Decomposition (Tasks 1 & 2)

[cite_start]The initial composite signal demonstrates a complex periodic structure resulting from the superposition of four cosine waves[cite: 17, 21, 23]. Figure ??

illustrates the high-frequency ripples superimposed on the primary low-frequency oscillation.

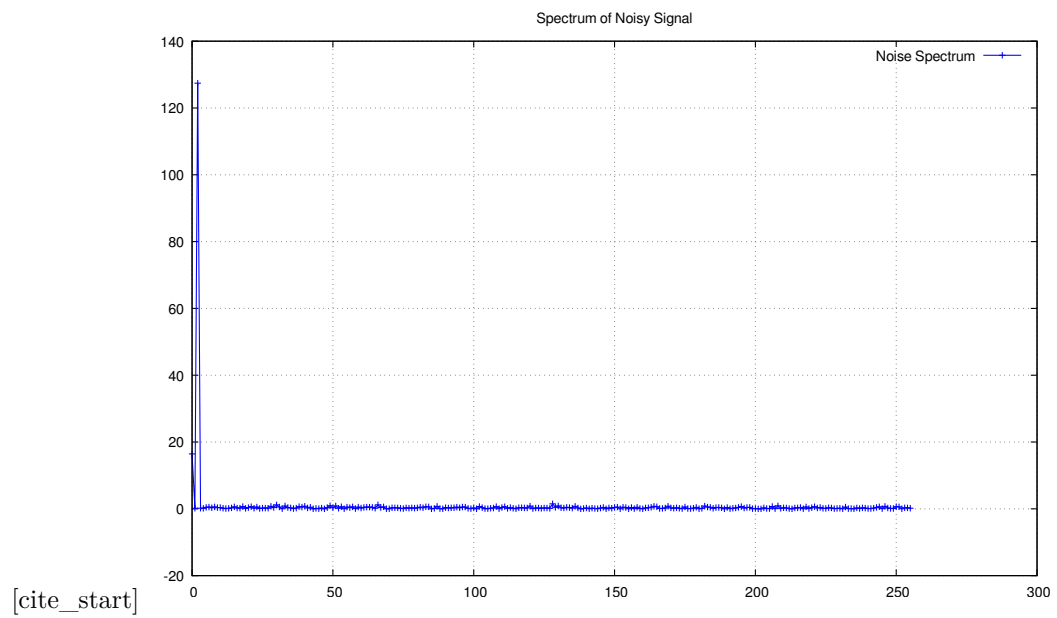
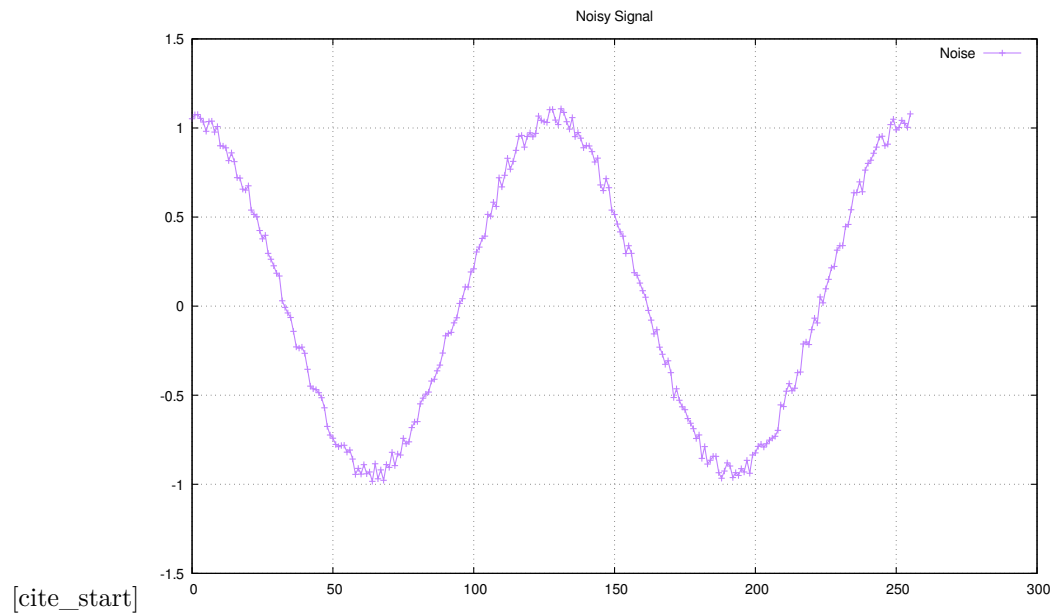


The frequency spectrum obtained via FFT (Figure ??) clearly identifies the constituent components. [cite_start]Four distinct peaks are visible at indices corresponding to the input frequencies ($k = 2, 8, 16, 64$), with magnitudes of approximately 128.0, 25.6, 16.0, and 8.0, respectively, which align with the theoretical amplitudes[cite: 17, 21, 24].

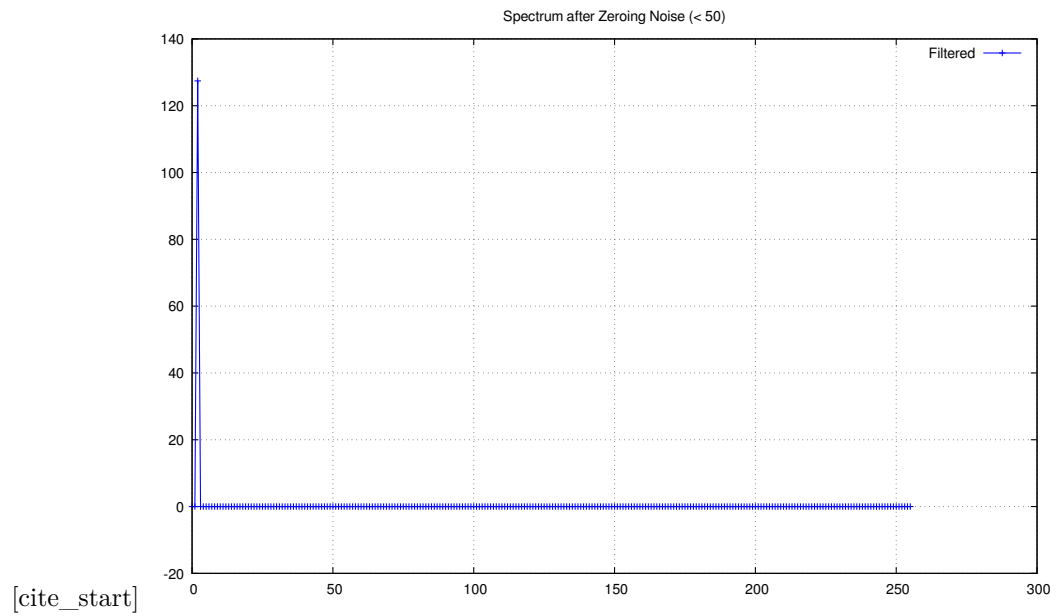


3.2 Noise Remediation (Tasks 3, 4, & 5)

[cite_start]The introduction of random white noise produced a jagged time-domain function, partially obscuring the underlying wave structure (Figure ??)[cite: 17, 21, 25]. [cite_start]In the frequency domain, this interference manifested as a broad noise floor distributed across nearly the entire spectral range (Figure ??)[cite: 17, 21, 26].

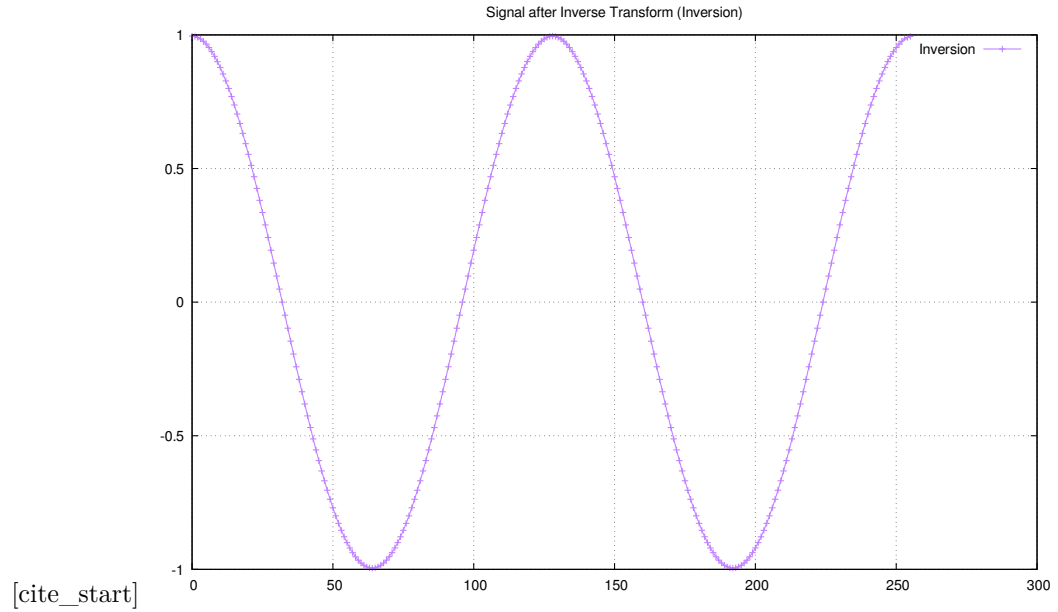


[cite_start]By applying a magnitude threshold of 50, the noise floor was eliminated, leaving only the primary signal peak intact (Figure ??)[cite: 17, 21, 19].



3.3 Reconstructed Signal (Task 6)

[cite_start]The reconstructed signal after inverse FFT (Figure ??) is significantly smoothed compared to the noisy input[cite: 17, 21, 20]. [cite_start]The periodicity is recovered, and the function is confined to the expected $[-1, 1]$ amplitude range, demonstrating successful noise removal through frequency-domain filtering[cite: 17, 21].



4 Conclusion

The laboratory results confirm that the Fast Fourier Transform is a robust method for analyzing multi-component signals and implementing noise-reduction filters [1]. By utilizing adaptive spectral thresholding, stochastic interference can be effectively isolated from the primary data [2]. Reconstructing the filtered signal confirms that frequency-domain manipulation preserves the essential periodic characteristics of the original wave while removing high-frequency disturbances.

Bibliography

- [1] GNU Project: Fast fourier transforms (ffts) in gsl. <https://www.gnu.org/software/gsl/doc/html/fft.html> (2024), accessed: 2024
- [2] Ważniak MIMUW: Szybka transformata fouriera. <http://wazniak.mimuw.edu.pl/index.php?title=MN10> (2024), theoretical background for FFT